

Faculty of Science, Engineering and Built Environment

SEE308 Electrical Machines and Drives

Deakin University Unit Guide

Trimester 2, 2025

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Welcome

This unit aims to provide students with principles of magnetic circuits, transformers, characteristics of electrical machines and drives with their power electronics fundamentals which are all used for many industrial applications such as utility power system, power plants, industrial processes as well as renewable energy systems such as solar and wind power generation and electric vehicle applications. This unit will also introduce variable speed motor drives and associated power electronic converters. This unit helps develop advanced knowledge and hands-on experience in designing power system components to ensure reliable and efficient network operation with the integration of renewable energy sources.
This Unit Guide provides you with the key information about this unit. Please read it carefully and refer to it frequently throughout the study period. Your unit site also provides information about your rights and responsibilities. We will assume you have read this before the unit commences, and we expect you to refer to it throughout the study period.

To be successful in this unit, you must:

- read all materials in preparation for your learning activities and follow up each with further study and research on the topic
- start your assessment tasks well ahead of the due date
- read or listen to all feedback carefully and use it in your future work
- attend and engage in all educator facilitated (scheduled) learning activities and other learning experiences as part of the unit design

Who is the unit team?

Unit chair: leads the teaching team and is responsible for overall delivery of this unit

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Other members of the team and how to contact them

Name: Associate Professor Enamul Haque, Lecturer
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Administrative queries

- check-out the 'SEBE Student Hub' section on your unit site
- contact your Unit Chair or Campus Leader
- drop in or contact Student Central to speak with a Student Adviser

For additional support information, please see the Rights and Responsibilities section under 'Content' in your unit site.
About this unit

Unit development in response to student feedback

Every trimester, we ask students to tell us, through eVALUate, what helped and hindered their learning in each Unit. You are strongly encouraged to provide constructive feedback for this Unit when eVALUate opens (you will be emailed a link).
In previous versions of this unit, students have told us that these aspects of the unit have helped them to achieve the learning outcomes:

- Practical activities and lab work
- Content closely linked with the real-world problems and industrial challenges

The following aspects of the unit have been introduced, enhanced or retained in response to feedback from students who have undertaken this unit in previous trimesters:

- Lecture material has been improved with recorded lectures.
- More support for Matlab/Simulink application will be provided.
- Timely responses to student queries and feedback will be provided.

If you have any concerns about the Unit during the trimester, please contact the unit teaching team - preferably early in the trimester - so we can discuss your concerns, and make adjustments, if appropriate.

Learning Outcomes

Each Unit in your course is a building block towards Deakin's Graduate Learning Outcomes - not all Units develop and assess every Graduate Learning Outcome (GLO).

ULO	These are the Unit Learning Outcomes (ULOs) for this unit. At the completion of this unit, successful students can:	Alignment to Deakin Graduate Learning Outcomes (GLOs)
ULO1	Apply fundamental technical knowledge to demonstrate the understanding of operation principles and characteristics of different electrical machines and drives.	GLO1: Discipline-specific knowledge and capabilities GLO4: Critical thinking
ULO2	Demonstrate the application of systematic approaches to select appropriate machine for different industrial applications as well as the control design for the selected machine.	GLO1: Discipline-specific knowledge and capabilities GLO3: Digital literacy GLO4: Critical thinking GLO5: Problem solving
ULO3	Apply problem-solving and research principles as part of the design of electrical machines and drive system.	GLO3: Digital literacy GLO5: Problem solving
ULO4	Demonstrate creative and innovative solutions to engineering problems associated with electrical machines and drives.	GLO4: Critical thinking GLO5: Problem solving
ULO5	Enhance communication and writing skills through active engagement during practical/project work in a team environment and reports on practical findings.	GLO2: Communication GLO6: Self-management GLO7: Teamwork

Assessing your achievement of the unit learning outcomes

Hurdle requirements

To be eligible to obtain a pass in this unit, students must achieve a mark of at least 40% in the problem solving online tests (AT1) and a mark of at least 50% in the Team project report (AT3).

Brief summary of the hurdle requirements	Rationale
To be eligible to pass in this unit, student must • achieve a mark of at least 40% in the Online Tests (AT1) • achieve a mark of at least 50% in the Team Project Report (AT3) This unit-specific activity may be undertaken either at scheduled session during the trimester (for on campus students) or during the unit-specific activities scheduled during the 'intensive' week (for online students).	The students are required to fulfil the hurdle requirements to achieve the unit learning outcomes (ULOs) and course learning outcomes (CLOs)

Summative assessment (tasks that will be graded or marked)

NOTE: It is *your responsibility* to keep a backup copy of every assignment and the materials used to develop/complete it where possible (e.g. written/digital reports, essays, videos, images). In the unusual event that one of your submissions becomes corrupted, is incorrectly submitted or otherwise lost, you may be asked to submit the backup copy. Any work you submit may be checked by electronic or other means for the purposes of detecting breaches of academic integrity such as collusion, plagiarism and contract cheating. You must understand your responsibility to act with honesty and

integrity in your studies as Deakin takes all breaches very seriously. Make sure you read Your rights and responsibilities as a student in this unit to find out more about academic integrity.

Deakin has a universal assessment submission time of 8 pm AEDT/AEST. A late penalty will apply to assessments submitted after 11.59 pm AEDT/AEST.

Summative assessment task 1

Title	**Problem solving online tests**
Brief description of assessment task	This assessment task requires students to demonstrate their understanding of the key terminology and concepts in electrical machines and drives, including their design and control mechanisms. The problem solving tests will be set at two points during the trimester.
Detail of student output	This is an individual task. Students will be required to complete written answers and mathematical calculations in response to a series of problem scenarios within a specific time frame. Students are expected to demonstrate their problem solving and critical thinking skills to develop relevant designs and analyse the operation of electric motors and drives.
Grading and weighting (% total mark for unit)	50% (2 x 25%) - marked and graded
This task assesses your achievement of these Unit Learning Outcome(s)	After completion of this assessment task, students will be able to demonstrate/achieve the following ULOs: **ULO1** - Demonstrate application of fundamental technical knowledge to demonstrate the understanding of operation principles and characteristics of different electrical machines and drives. **ULO3** - Demonstrate application of problem-solving and research principles as part of the design of electrical machines and drive system.
This task assesses your achievement of these Graduate Learning Outcome(s)	**GLO1** - Discipline knowledge and capabilities: appropriate to the level of study related to a discipline or profession. **GLO3** - Digital literacy: using technologies to find, use and disseminate information. **GLO4** - Critical thinking: evaluating information using critical and analytical thinking and judgement. **GLO5** - Problem solving: creating solutions to authentic (real world and ill-defined) problems.
How and when you will receive feedback on your work	During the seminar and online sessions, students will work through a range of problem scenarios similar to that in this assessment task. Students will be able to identify the gaps in their understanding and place efforts where required. Students will receive written feedback on their results within 15 days via unit site. Students are also encouraged to contact the unit chair if additional support is required prior to the test.
When and how to submit your work	**Week 5 and Week 11.** Students will submit the answers via the unit site. **Test 1: Week 5:** Friday 8 August 2025 at 8pm (AEST) **Test 2: Week 11**: Thursday 25 September 2025 at 8pm (AEST) Time allowed: 2 hours plus 15 minutes for scanning and uploading the test paper.

Summative assessment task 2

Title	**Laboratory reports**
Brief description of assessment task	Practical activities and final written reports. Students will be completing laboratory experiments on connecting DC motors and controlling their operation as well as synchronous and asynchronous motors. The findings from these experiments will be presented in the form of two professionally written laboratory reports. Students are expected to demonstrate their critical thinking skills to analyse and interpret the laboratory results. Further details will be provided on the unit site.

Detail of student output	Students will be required to write two laboratory reports based on their practical activities (and supporting guidelines), and submit it via the unit site. The first report will comprise the findings from the first three laboratory experiments and the second report will collate the results from the remaining laboratory experiments. This is an individual assessment task.
Grading and weighting (% total mark for unit)	25% (15% for Report 1 and 10% for Report 2) - Marked
This task assesses your achievement of these Unit Learning Outcome(s)	After completion of this assessment task, students will be able to demonstrate/achieve the following ULOs: **ULO1: Apply** fundamental technical knowledge to demonstrate the understanding of operation principles and characteristics of different electrical machines. **ULO2: Demonstrate** the application of systematic approaches to select appropriate machine for different industrial applications as well as the control design for the selected machine. **ULO3: Apply** problem-solving and research principles as part of the design of electrical machines and drive. **ULO4: Demonstrate** creative and innovative solutions to engineering problems associated with electrical machines. **ULO5: Enhance communication and writing skills** through active engagement during practical/project work in a team environment and reports on practical findings**.**
This task assesses your achievement of these Graduate Learning Outcome(s)	After completion of this assessment task, students will be able to demonstrate/achieve the following GLOs: **GLO1:** By demonstrating the ability to explain and apply their understanding of the concepts and technical knowledge in electrical machines. **GLO2:** By demonstrating the ability to communicate results/data from experimental investigations to professional and non-professional stakeholders. **GLO4:** By demonstrating the ability to critically analyse experimental results. **GLO5:** By demonstrating the ability to solve problems related to electrical machines. **GLO6:** By demonstrating the ability to reflect on their learning during laboratory session through their individual laboratory reports. **GLO7:** By demonstrating the ability to work in a team environment during laboratory sessions.
How and when you will receive feedback on your work	Students will be given instructions/ feedback during the trimester through the unit site and/or during lab sessions.
When and how to submit your work	Students are required to submit practical reports as follows: **Campus Students:** • **Report 1** Sunday of **Week 6** (24 August 2025, 8pm AEST) • **Report 2** Sunday of **Week 9** (14 September 2025, 8pm AEST) **Online Students:** • **Report 1 and 2** Sunday of **Week 9** (14 September 2025, 8pm AEST) Online submission via the unit site.

Summative assessment task 3

Title	**Team project report**
Brief description of assessment task	This assessment evaluates the students understanding of fundamental knowledge related to electrical machines and drives learnt during the trimester. Students are required to clearly and professionally present their project outcome in written technical report which include final project details, design specifications for different motors, associate computer models and simulation results of their operational characteristics.

	Students will be given a project to complete in a team during the trimester. The students will form a project team of up to four members and provide the innovative solutions for the given problem, which reflects their conceptual understandings of electrical machine topics. Further details will be given in the unit site.
Detail of student output	This is a team based assessment task, however, there are some individual components as outlined below. Each team member will contribute to the final report (30 pages maximum). Students are required to: • work in a group of 4 students as one team and provide the solutions for the given problem, demonstrate their knowledge, understanding, learning experiences and writing skills about the project work related to electrical machines and drive system; and • demonstrate the ability to work in a team environment. The report will include: • Background, Project objectives and outcomes • Literature review and theoretical background (individual) • Design and theoretical analysis • Computer simulation models • Discussions of simulation results (Individual) • Learning experiences, suggested recommendations for improvement, reflection on their teamwork skills.
Grading and weighting (% total mark for unit)	25% (graded)
This task assesses your achievement of these Unit Learning Outcome(s)	After completion of this assessment task, students will be able to demonstrate/achieve the following ULOs: **ULO1: Apply** fundamental technical knowledge to demonstrate the understanding of operation principles and characteristics of different electrical machines and drives. **ULO2: Demonstrate** the application of systematic approaches to select appropriate machine for different industrial applications as well as the control design for the selected machine. **ULO3: Apply** problem-solving and research principles as part of the design of electrical machines and drive system. **ULO4: Demonstrate** creative and innovative solutions to engineering problems associated with electrical machines and drives. **ULO5: Enhance communication and writing skills** through active engagement during practical/project work in a team environment and reports on practical findings**.**
This task assesses your achievement of these Graduate Learning Outcome(s)	After completion of this assessment task, students will be able to demonstrate/achieve the following GLOs: **GLO1:** By demonstrating the ability to explain and apply their understanding of the concepts and technical knowledge in electrical machines and drives. **GLO3:** By demonstrating the ability to collect information from online resources and use industry standards to conduct the project work. **GLO4:** By demonstrating the ability to critically analyse results and findings. **GLO6:** By demonstrating the ability to work and learn independently and reflect on their learning during project work through their project report. **GLO7:** By demonstrating the ability to work in a team environment by working in team project.
How and when you will receive feedback on your work	Feedback will be provided progressively during the project design phase and after submission.
When and how to submit your work	Final project report is to be submitted by the end of **Week 10** (on Sunday 21 September 2025 at 8pm AEST) via the unit site.

Your learning experiences in this unit

Educator-facilitated (scheduled) learning activities - on-campus unit enrolment

1 x 1 hour seminar per week, 1 x 2 hour practical experience (laboratory) per week.
Educator-facilitated (scheduled) learning activities - online unit enrolment

Online independent and collaborative learning including a 1 x 1 hour seminar per week. Students are required to attend and participate in project and practical experience (laboratory) based activities at the Waurn Ponds (Geelong) Campus for the scheduled sessions during the trimester intensive activities as detailed in the unit site.
Typical study commitment

Students will on average spend 150 hours over the teaching period undertaking the teaching, learning and assessment activities for this unit.
This will include educator guided online learning activities within the unit site.
For campus students learning and assessment activities include watching recorded lectures, attending designated laboratory, seminars and project activities, assessment tasks, readings and study time. For online students the time should be spent watching recorded lectures, online learning activities, designated laboratory, seminar and project activities, assessment tasks, readings and study time.

The unit site will host all the unit materials including lecture notes, laboratory manual, reading materials and assessment details.
Note

At Deakin, courses are delivered within a learning environment that provides all students with equitable and consistent access to facilities, infrastructure, resources, and support to assist student progress and achievement of learning outcomes.

We have introduced new terms to reflect learning activities to enhance your learning experience, aligning with our innovative **DeakinDesign learning principles and practices**. The new terms better reflect how teaching teams will guide you through your learning journey and the types of learning experiences you will have.

'Lectures' are the activities where teaching staff engage you through presentations with student participation.

In **'seminars'**, an educator will guide you in a smaller group of students through highly interactive discussions and activities.

Your units may also include '**practical experiences**' such as '**laboratory**', '**workshops**', '**clinical skills**' and more. These hands-on activities typically take place in specialised facilities with industry tools, equipment or technology to allow you to apply your knowledge practically.

Some other terms

If you see a '**meeting**' in your timetable, this is an optional drop-in session.

'Assessments' or **'team-based learning**' indicate an in-class evaluation of your skills or knowledge. A **'pre-assessment practice'** could be scheduled to prepare you for these assessments.

Find out more

Take a look at the **Learning activities webpage** for a full list of the terminology changes and reasons they were changed.
Unit learning resources

Your unit learning resources can be accessed from your unit site.
The texts and reading list for SEE308 can be found via the University Library.

Note: Select the relevant trimester reading list. Please note that a future teaching period's reading list may not be available until a month prior to the start of that teaching period so you may wish to use the relevant trimester's prior year reading list as a guide only.
Essential learning resources

- There is no prescribed text book for this unit, however, there are some recommended books for this unit (see below).
- Detailed lecture notes will be provided. All the required materials will be uploaded in the unit site in due time.

Recommended learning resources

The Deakin Software Library provides students with access to software that you may need or find useful for your study at Deakin.

Recommended texts:

- Electric Machinery Fundamentals, Author: Stephen J. Chapman, Edition: 5th, ISBN-10: 0073529540, ISBN-13: 978-0073529547
- P. C. Sen, "Principles of Electric Machines and Power Electronics" 3rd Edition, Wiley, 2014

Recommended references:
There are a wide variety of textbooks and internet resources useful for this unit. Some of the main resources will be introduced to students in the first week.
Where to access unit resources

Textbooks can be sourced from various outlets including direct from the publisher, online bookshops, or retailers. Limited copies of textbooks are also available on loan from the University Library.
Key dates for this study period

Trimester 2 teaching period begins	Monday 7 July 2025
Census Date	Friday 15 August 2025
Intra-trimester break	Monday 11 August - Sunday 17 August 2025
Trimester 2 teaching period ends	Friday 26 September 2025
Study period (end-of-unit assessment/examination preparation period)	Monday 29 September - Friday 3 October 2025
End-of-unit assessment/examination period	Monday 6 October - Friday 17 October 2025
Inter-trimester break (the period between trimesters)	Monday 20 October - Friday 31 October 2025
Unit results released	Friday 31 October 2025 (10.30 am)

Unit weekly activities

Week	Commencing	Topic	Special learning activities	Assessment due date
1	7 July 2025	Magnetic circuits	Lab induction	
2	14 July 2025	DC Machines Part 1	Lab 1 (software)	
3	21 July 2025	DC Machines Part 2	Lab 2	
4	28 July 2025	DC Motor Drives	Lab 3	
5^	4 August 2025	Synchronous Machines Part 1	Project discussion/Tutorial	**Assessment 1: Online test 1** due 8 August
6	18 August 2025	Synchronous Machines Part 2	Intensive Week (online students)	**Assessment 2: Laboratory report 1 (Campus students)** due 24 August
7	25 August 2025	Induction Machines Part 1	Lab 4	
8	1 September 2025	Induction Machines Part 2	Lab 5	

Week	Commencing	Topic	Special learning activities	Assessment due date
9	8 September 2025	AC Machine Drive Part 1	Project discussion/Tutorial	**Assessment 2: Laboratory report 1 (Campus students)** due 14 September **Assessment 2: Laboratory report 1 and 2 (Online students)** due 14 September
10	15 September 2025	AC Machine Drive Part 2	Project discussion/Tutorial	**Assessment 3: Team project report** due 21 September
11#	22 September 2025	Review	Tutorial/Review	**Assessment 1: Online test 2** due 25 September
12	29 September			Study period
13-14	6 October			End-of-unit assessment period

^ Intra-trimester break: **Monday 11 August – Sunday 17 August 2025**
AFL Grand Final Eve public holiday: **Friday 26 September 2025** - University closed